

Yonghan Yang

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Objective

I'm an **aspiring AI researcher**, focused on Score-Based Generative Models, Agentic AI and Foundation Models. I'm also interested in interdisciplinary fields: AIDD in biology and causal inference in finance. I sketch, play tennis, enjoy literature, run regularly and volunteered across venues. My motto is *To become a Hero the Hard Way*.

Education

MBZUAI, (Abu Dhabi, UAE), *GPA - 3.8/4* Aug. 2025 – Present

- Bachelor of Science (B.Sc.) in Artificial Intelligence, expected May 2029
- Courses: **ML7501 Applied ML** (M.S. course), AI1225 Algorithms & Data Structures, AI1120 Probability & Statistics, AI1100 Calculus & Linear Algebra, AI1010 Introduction to AI, AI1030 Python Programming, AI1220 Software, Web & Mobile Engineering, GE1510 Physics & Life Science in the Age of AI, BUS1010 Economics, UG0500 Advanced Math
- Honors: Dean's List (inaugural cohort, Fall 2025, 10%), *Sheikh Tahnoon bin Zayed Scholarship in AI Excellence*, Pioneer Scholarship

Beijing National Day School, (Beijing, China), *GPA - 4.2/4.3 UW* Sept. 2019 – July 2025

- High School Advanced Placement (AP) program jointly with [Wasatch Academy](#)
- Courses: Calculus III, Differential Equations, Linear Algebra, Probability Theory, Mathematical Modeling, Sociology, Marine Biology
- Tests: SAT 1550 (Math 800, English 750); TOEFL 111/120; CEFR C1 (191/200); 11 AP exams with eight 5s (Calculus BC, Physics C: Mechanics, Physics 1, Chemistry, Biology, Computer Science A, Statistics, Psychology)

Experience

Cyber-Physical Intelligence Lab & AAAAA Community, *Researcher* Oct. 2025 – Present

AI labs across MBZUAI, McGill University & [Mila](#). CPIL advances trustworthy, efficient, and agentic AI for the cyber-physical world; AAAAA Community is a research collective spanning universities, labs, and industry (MIT, University of Cambridge, Imperial College London, Salesforce, RBC Borealis, and more). Directly mentored by lab founder [Prof. Steve Liu](#) and senior member [Ye Yuan](#).

- **SPADE: Support-Proximity Augmented Diffusion Estimation for Offline Black-Box Optimization** (*co-first author*, Oct. 2025 – Jan. 2026): experimented with conditional diffusion models as surrogates for offline optimization, with calibration and a kNN support-proximity prior that keeps the optimizer out of unsupported regions; the regularizer is provably equivalent to Bayesian inference under a valid design prior. Mean rank 2.8/24 on Design-Bench. Accepted by 2026 [ICML](#) (poster presented in Seoul, Jul. 2026) and the [ICLR 2026 DeLTa workshop](#) (Deep Generative Model in Machine Learning: Theory, Principle and Efficacy). [[Code](#)] [[Project Page](#)] [[ICML](#)] [[OpenReview](#)] [[Video](#)] [[Poster](#)] [[Slides](#)] [[BibTeX](#)]
- **Surrogate-Guided Memory Retrieval for Agents** (*co-first author*, Apr. 2026 – Present): proposed and benchmarked surrogate-guided offline training of memory retrieval for autonomous agents, scoring each memory by how much it helps downstream tasks rather than by semantic similarity alone. Submitted to the 2026 NeurIPS Workshop on Dynamic Alignment in Human-AI Coupled Systems ([BiAlign](#)).
- **QUACK: Questioning, Understanding, and Auditing Communicated Knowledge in Multimodal Social Deduction Agents** (Feb. 2026 – Present): audited grounding of agents' claims in social deduction games with an open-source environment and evaluation framework that checks whether claims are grounded in what agents saw and did; even the strongest frontier VLM hallucinates 15.1% of its verifiable spatial claims. Accepted by 2026 [EMNLP](#) Main Conference. [[Code](#)] [[Dataset](#)]
- **Discrete Diffusion Models: A Unified Framework from Tokenization to Generation** (Mar. 2026 – Present): investigated AI for Science (AI4S) applications for a comprehensive survey that unifies the formulation and applications of discrete diffusion models in a single design space, from tokenization to generation. Under review at [TMLR](#). [[Paper List](#)]

- **Herculean: An Agentic Benchmark for Financial Intelligence** (Mar. 2026 – May 2026): tested agents on representative financial intelligence workflows across trading, hedging, market insights, and auditing; frontier agents do well at trading and market insights but struggle with hedging and auditing. Accepted by 2027 ACL. [Code] [Dataset]

Precision-medicine AI Lab, *Researcher*

Mar. 2026 – May 2026

AI4S lab across MBZUAI & Harvard Medical School, integrating multi-scale biomedical data (genomics, transcriptomics, electronic health records) for precision medicine. Directly mentored by lab founder Prof. Jun Wen.

- **Semi-supervised High-order Relation Learning**: trained a multi-head attention (MHA)-based model on a comprehensive drug combination-disease relationship dataset, predicting and analyzing high-order relations among drugs and diseases.

Paper Reviewer

Feb. 2026 – Present

Reviewed for top ML venues: the 2027 ICLR main conference and the 2026 ICLR 2nd Workshop on Foundation Models for Science (FM4Science).

GenBio AI, *Research Engineer Intern*

Mar. 2026 – Present

AI4S startup (Palo Alto, Paris & Abu Dhabi) co-founded by world-renowned scientists like Nobel laureate Prof. David Baker, building AIDO, an AI-Driven Digital Organism toward a world model of human biology. Supervised by CTO Prof. Le Song.

- **BioXArena: Benchmarking LLM Agents on Multi-Modal Biomedical Machine Learning Tasks**: designed multi-modal biomedical ML agent tasks and benchmarked agents, with 76 end-to-end tasks across 9 biomedical domains, hidden labels, and held-out graders; made key contributions to the formulation of the paper. Accepted by 2026 NeurIPS (Evaluations & Datasets Track, poster). [Code] [Project Page] [Dataset] [OpenReview]
- **VCHarness**: optimized the efficiency of the MCTS-based harness, an autonomous AI system that builds virtual-cell perturbation-response models with an AI coding agent and multimodal biological foundation models, and benchmarked it on BioXArena. [Project Page] [Paper] [Blog]

Shanghai Academy of AI for Science, *Research Engineer Intern*

May 2026 – Present

AI4S R&D institute focused on industrial applications, founded in 2023 by the Shanghai municipal government and Fudan University. Engineered the A2A (Agent2Agent) protocol and researched generative models.

- **Decentralized Agentic Reasoning**: optimized a decentralized agent swarm built on the established A2A protocol.

Publications

Support-Proximity Augmented Diffusion Estimation for Offline Black-Box Optimization

Yonghan Yang*, Ye Yuan*, Zipeng Sun, Linfeng Du, Bowei He, Haolun Wu, Can Chen, Xue Liu

ICML 2026, PMLR 306. [arXiv:2605.11246](#)

Discrete Diffusion Models: A Unified Framework from Tokenization to Generation

Ye Yuan, Weien Li, Rui Song, Zeyu Li, Haochen Liu, Xiangyu Kong, Zixuan Dong, Linfeng Du, Zipeng Sun, Weixu Zhang, Jiaxin Huang, Changjiang Han, Yonghan Yang, Zichen Zhao, Xiuyuan Hu, Haolun Wu, Yankai Chen, Fengran Mo, Jikun Kang, Bowei He, Dawn Song, Philip S. Yu, Xue Liu

Under review at TMLR. [arXiv:2607.13431](#)

QUACK: Questioning, Understanding, and Auditing Communicated Knowledge in Multimodal Social Deduction Agents

Ye Yuan, Rui Song, Weien Li, Zeyu Li, Haochen Liu, Xiangyu Kong, Changjiang Han, Yonghan Yang, Zichen Zhao, Zixuan Dong, Fuyuan Lyu, Bowei He, Haolun Wu, Jikun Kang, Xue Liu

EMNLP 2026. [arXiv:2605.27068](#)

BioXArena: Benchmarking LLM Agents on Multi-Modal Biomedical Machine Learning Tasks

Loka Li, Duzhen Zhang, Xingbo Du, Leonard Song, Zixiao Wang, Assanali Aukenov, Noel Thomas, Shakhnazar Sailaukan, Yonghan Yang, Feilong Chen, Jiahua Dong, Kun Zhang, Bin Zhang, Le Song

NeurIPS 2026. [arXiv:2605.15766](#)

Herculean: An Agentic Benchmark for Financial Intelligence

Xueqing Peng, Zhuohan Xie, Yupeng Cao, . . . , Haolun Wu, *Yonghan Yang*, Zichen Zhao, . . . , Jian-Yun Nie, Sophia Ananiadou (63 authors)

ACL 2027. [arXiv:2605.14355](#)

* Equal contribution.

Awards & Honors

- 2025 MBZUAI *Dean's List* (inaugural cohort, Fall 2025, 10%).
- 2025 MBZUAI *Sheikh Tahnoon bin Zayed Scholarship in AI Excellence & Pioneer Scholarship*.
- 2025 IMMC Greater China round *Outstanding* (< 1%) & International round *Meritorious* (< 10%).
- iGEM high school section *Gold Medal & Top 10* for three years (2023–2025); 2025 *Best Measurement & Best Part Collection* as instructor; 2024 *Best Wiki* nominee as student leader.
- 2024 John Locke Institute's Global Essay Prize *Finalist & Merit* in Economics category, Question 1.
- 2024 Goi Peace Foundation's International Essay Contest for Young People *Finalist* (top 100 of 10,233).
- 2024 ARML *National Silver*.
- 2024 Purple Comet! Math Meet team math competition China *2nd place* (global 16th).
- 2024 Johns Hopkins Math Tournament Team *Silver*, Power & Relay Round *Bronze*.
- UKBC's 2022 Intermediate Biology Olympiad *Bronze* & 2023 British Biology Olympiad *Global Gold*.
- 2023 COMAP's HiMCM *Honorable Mention*.
- 2022 Junior Canadian Chemistry Olympiad China *Gold* & Global *Bronze*.
- 2022 Berkeley Math Tournament *Honorable Mention* in Geometry Round & Team Overall.
- 2022 China Thinks Big Innovation Program research paper accepted to Digital Library Collection.
- AP Scholar Award (2023) & with Distinction (2024, 2025).
- 2019 FIRST Tech Challenge Beijing regional *first prize*.

Activities

Math Modeling Competitions, *COMAP* & *MAA* & *AMS* Jan. 2025 – Present

- 2025 IMMC *Greater China* (**Outstanding**, < 1%) and *International* (**Meritorious**, < 10%) rounds as *team leader*.
- Developed MILP modeling frameworks for two scheduling problems with PuLP library. Drafted and typeset both papers in $\LaTeX$.

Synthetic Biology, *iGEM Competition* & *Team Wiki* Oct. 2022 – Present

- Contributed to team BNDS-China as *instructor* (2025-2026), *student leader* (2024) and member (2023). Won **top 10** teams (10%) & gold medal in 2023-2025, plus Best Measurement & Best Part Collection in 2025. See wikis for [2023](#), [2024](#), [2025](#).
- 2024, as student leader: coordinated teammates and developed a programmable probiotic platform to detect, monitor, and regulate gut metabolites to help treat or prevent gut dysbiosis. Oversaw inducible promoter characterization; modeled genetic circuit expression dynamics and used protein structure prediction in dry lab.
- 2025, as instructor: supervised lab and outreach activities, and developed computational biology solutions for a programmable probiotic platform for feline gut health. Instructing the 2026 team (ongoing).
- Interviewed doctors & synthetic biologists. Organized conferences with 10+ teams and planned the 20-person trip to the Paris Jamboree.
- Introduced synbio to high school students as club president, prepared *agar art* for exhibition & formed a band during 2024 school music festival.

Rhino-Bird Science Talent Advancement Program, *BNRist* & *THU* & *Tencent* Apr. 2024 – Aug. 2024

- Learned research basics like drafting paper, calculus, probability theory, linear algebra, and CS-related knowledge like AI ethics, computer vision, LLM construction.
- Realized attention mechanism via PyTorch. Researched on dynamic game theory in college application.

- Completed & passed the 30 h Basic Science Research Training and course assessment.

Jingwan Applied Math Summer Camp, *Team Leader*

July 2023 – Aug. 2023

- Hosted by Beijing Normal University Mathematical Modeling Education Center. Learned basic math structures, data analysis techniques, and applied math methodologies.
- Modeled colony collapse disorder & analyzed bee hive data, leading team of three.

School Clubs, *Beijing National Day School*

Sept. 2022 – July 2025

- President of biology club *BioCamp* (school honor-level club, 2024): reviewed biology & chemistry as TA, taught biology olympiad, organized anatomical experiments and forest-park educational tours.
- Cofounder of math club *MathCorner*; vice president of origami club; member of Moonlight Discussion Club, Math Modeling Society and school badminton team (BASE Cup men's doubles quarter-finalist).

Volunteering

Sept. 2023 – Present

- Student volunteer at the *MBZUAI HCI Symposium 2025* (Nov.–Dec. 2025), supporting event operations for attendees, speakers, and organizers of a symposium on human-computer interaction and human-centered AI.
- Student volunteer at the *IUCN World Conservation Congress 2025* at ADNEC, Abu Dhabi (Sept.–Oct. 2025).
- Beijing ecological & environmental protection volunteer (2024); cameraman & volunteer at Laotzu College tea ceremonies.
- Cofounded a dyslexia support group in the local community: interviewed dyslexic people, wrote articles, and coauthored a 20-page illustrated leaflet.

First Tech Challenge, *School Team Programmer*

Oct. 2019 – Jan. 2020

- Learned Java grammar & OpenCV as the only programmer in 1 month. Programmed robot to recognize targets automatically, which won Beijing regional first prize.

Interests & Skills

- *Coding* in Python, Rust, TypeScript; deep learning with PyTorch. *Writing* via Markdown, Typst, $\text{\LaTeX}$. *Visuals* with Manim, PowerPoint, Photoshop.
- *AI tooling*: Anthropic certificates (Jan. 2026) in Building with the Claude API, Claude Code in Action, Model Context Protocol (Introduction & Advanced Topics), Claude 101, AI Fluency: Framework & Foundations, and AI Fluency for Educators; AWS Artificial Intelligence Practitioner Learning Plan (Apr. 2026).
- *Lab*: designed DNA sequences via SnapGene & Benchling, predicted protein structure via AlphaFold, analyzed binding via AutoDock Vina, and visualized structure via PyMOL.
- Bilingual in Mandarin & English. Learning French & Japanese.
- Sketch, play tennis, run 70km+ monthly; played foil & saber at local clubs and badminton tournaments on high school team; practiced origami & oil painting.